Project Report: Student Grade Calculator

1. Introduction
The Student Grade Calculator is a Python program designed to assign student grades based on marks obtained out of 100.

2. Objective
To develop a simple program that uses conditional statements to determine grades and validates user input.

3. Problem Statement
Manual grade assignment can be repetitive. This project automates grade assignment using predefined conditions.

4. Methodology
1. Accept the student's name.
2. Request marks out of 100.
3. Validate the input.
4. Assign a grade using conditional statements.
5. Display the student's details and result.

5. Grading Logic
- 90–100: A
- 80–less than 90: B
- 70–less than 80: C
- 60–less than 70: D
- Below 60: F

6. Technologies
Python 3, Visual Studio Code, and GitHub.

7. Testing
Test the program using valid marks, grade boundaries, negative marks, marks above 100, and non-numeric inputs. Record actual results after testing.

8. Learning Outcomes
- Python functions
- Conditional statements
- Loops
- Exception handling
- Input validation
- Testing and documentation

9. Conclusion
The project demonstrates how Python can be used to automate a simple grading task consistently.

10. Project Links
GitHub Repository: https://github.com/venky1318/student-grade-calculator